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$$\int \frac{1}{2+\cos x} dx \overset{t=\tan\left(\frac{x}{2}\right)}{=} \int \frac{1}{2+\dfrac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt = 2\int \frac{1}{2\left(1+t^2\right)+1-t^2} dt$$

$$= 2\int \frac{1}{t^2+3} dt$$

$$= \frac{2}{\sqrt{3}} \operatorname{Bgtan}\left(\frac{t}{\sqrt{3}}\right) + c \overset{t=\tan\left(\frac{x}{2}\right)}{=} = \frac{2}{\sqrt{3}} \operatorname{Bgtan}\left(\frac{\tan\left(\dfrac{x}{2}\right)}{\sqrt{3}}\right) + c$$

References